

UNIVERSITY COLLEGE LONDON

EXAMINATION FOR INTERNAL STUDENTS

MODULE CODE : **COMP0070**

ASSESSMENT PATTERN: **COMP0070A7PG 001**

MODULE NAME : **COMP0070 - Algorithmics**

LEVEL : **Postgraduate**

DATE: : **21-May-2025**

TIME : **14:30**

DURATION : **2 Hours**

This paper is suitable for candidates who attended classes for this module in the following academic year(s):

Year
2024-25

EXAMINATION PAPER CANNOT BE REMOVED FROM THE EXAM HALL. PLACE EXAM PAPER AND ALL COMPLETED SCRIPTS INSIDE THE EXAMINATION ENVELOPE

Hall instructions	N/A
Additional materials	None
Standard Calculators	Yes
Non-Standard Calculators	No

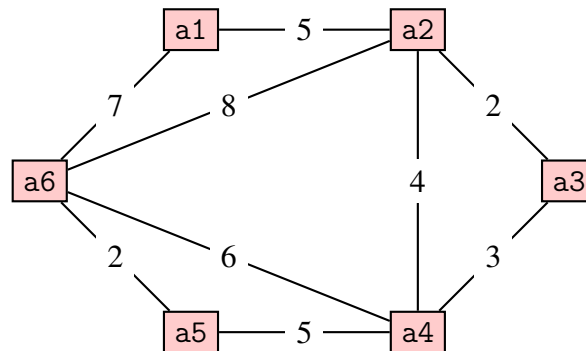
TURN OVER

Answer BOTH questions.

Marks for each part of each question are indicated in square brackets

Calculators are permitted

1. a. Consider the following undirected graph. Suppose we use $a1$ as the arbitrary starting node. Apply Prim's algorithm to find a minimum spanning tree. Give this minimum spanning tree. Also give intermediate values for the Nearest and MinDist arrays obtained with each cycle of the algorithm.



[12 marks]

- b. If depth-first search is undertaken on a binary tree starting at the root, is it the same as any of preorder traversal, inorder traversal, or postorder traversal? If so, which? Give a precise and clear explanation with your answer. The answer should include appropriate definitions.

[6 marks]

- c. This question is about the greedy approach to finding a simple cycle involving all the nodes in a graph. This is referred to as the travelling salesperson problem.

- Provide an explanation for how a greedy approach could be designed for this task. It is not necessary to give pseudocode, but do clearly explain how the greedy choice works in trying to obtain an optimal solution.
- Give a weighted undirected graph with 5 nodes and 10 arcs for which the greedy approach gives an optimal solution.

[10 marks]

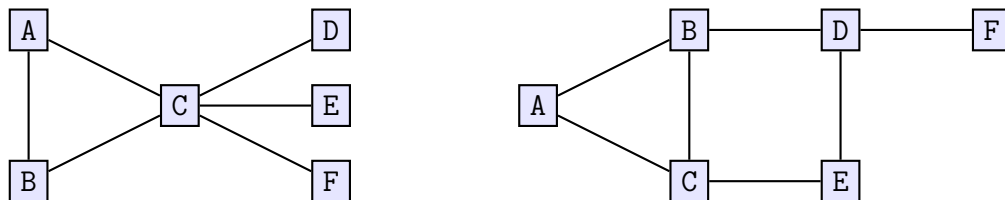
- d. Let the length of a branch of a tree be the number of arcs traversed when travelling from the root to the leaf of the branch. Let the height of a tree be the length of the longest branch of the tree. Using this definition for height, what is the height of a heap with n elements? Give a clear, precise, and concise explanation with your answer.

[6 marks]

- e. A binary tree can be encoded using two functions l and r such that for a node n , $l(n)$ gives the left child of n (or nil if there is none) and $r(n)$ gives the right child (or nil if there is none). Let $\text{Height}(l, r, x)$ be a simple recursive algorithm for taking a binary tree encoded by the l and r functions together with the root node x for the binary tree, and returns the height of the binary tree. Give pseudocode for this algorithm. Briefly explain your answer.

[10 marks]

- f. A **node cover** of a graph $G = (N, A)$ is a set of nodes $C \subseteq N$ that includes at least one endpoint of every arc of the graph (i.e. for each $(x, y) \in A$, $x \in C$ or $y \in C$). A **minimal node cover** of G is node cover of G with a minimal number of nodes (i.e. C is a minimal node cover of G if there is no node cover C' of G such that $|C'| < |C|$). A practical situation of using this is an organization wanting to install the fewest possible closed circuit cameras covering all hallways (arcs) connecting all rooms (nodes) on a floor. For each of the following two graphs, give a minimal cover and a non-minimal cover.



[6 marks]

2. a. i. Succinctly explain the purpose of big-O notation.

[1 marks]

- ii. State the definition of big-O notation.

[2 marks]

- iii. Using the definition, justify whether the following statements are true or false.

$\alpha. \quad n \cdot \log n \quad \in \quad O(n^2)$

[2 marks]

$\beta. \quad 3^{n+3} \quad \in \quad O(3^n)$

[2 marks]

$\gamma. \quad 3^n \quad \in \quad O(3^{n+3})$

[2 marks]

$\delta. \quad 3^{3n} \quad \in \quad O(3^n)$

[2 marks]

$\epsilon. \quad 3^n \quad \in \quad O(3^{3n})$

[2 marks]

$\zeta. \quad \log(n \cdot \log n) \quad \in \quad O(\log n)$

[2 marks]

- b. The function funny(n), shown below, takes a positive integer argument n and returns a nonnegative integer c.

```
funny ( n ) :  
    m := 1  
    c := 0  
    while m < n :  
        m := m + m  
        c := c + 1  
    return c
```

- i. Explain, precisely yet succinctly, what this function does, *i.e.*, the relationship between n at the beginning and c at the end.

[4 marks]

- ii. Write an expression that gives the exact running time of this function, assuming that the elementary operation of interest is addition +; your expression should be a function of the input parameter n .
[3 marks]
 - iii. Argue succinctly why your answer to (2.b.ii) above is correct.
[3 marks]
 - iv. Use big-O notation to restate your answer to (2.b.ii) above.
[1 marks]
 - v. Write an expression that gives the exact running time of the funny function, assuming that the elementary operations of interest are addition +, comparison $<$, and assignment $:=$.
[2 marks]
 - vi. Use big-O notation to restate your answer to (2.b.v) above.
[1 marks]
- c. Give the runtime for each of the following algorithms using big-O notation.
- i. Insertion sort
[2 marks]
 - ii. Merge sort
[2 marks]
 - iii. Quicksort, assuming worst-case pivot selection
[2 marks]
 - iv. Quickselect, assuming optimum pivot selection
[2 marks]

d. Consider the recurrence relation

$$T(n) = \begin{cases} T(\lfloor \sqrt{n} \rfloor) + 1 & \text{when } n > 1 \\ 0 & \text{otherwise} \end{cases}$$

N.B. The floor function $\lfloor x \rfloor$ represents rounding x down, *e.g.* $\lfloor 3.14 \rfloor = 3$.

i. Calculate the solution to $T(n)$ for the following values of n : 2 , $2^2 = 4$, $4^2 = 16$, $16^2 = 256$, and $256^2 = 65,536$.

[2 marks]

ii. Devise a closed-form expression (as a function of n) that solves the above recurrence relation. *Hint*: you might find it useful to express the sequence given above in (2.d.i) in a more uniform format.

[4 marks]

e. i. Give an example of a problem in P.

[1 marks]

ii. Give an example of a problem in NP.

[1 marks]

iii. Give an example of a problem in NP-COMPLETE.

[1 marks]

iv. Suppose $X \leq_P Y$ and $X \in P$. What do we know about Y ?

[2 marks]

v. Suppose $X \leq_P Y$ and $Y \in P$. What do we know about X ?

[2 marks]

END OF PAPER